Highlights

- One LLM request per paper yields gene–disease pairs with checkable evidence.
- Four statuses separate supported, hedged, negative and absent relations.
- Precision 0.894 on 419 BioRED abstracts; few contradictions of negative labels.
- Re-querying missed pairs added no relations: the gap is an evidence standard.
- A reason-code operating point raised held-out F1 from 0.542 to 0.627.

Statement of significance

[TODO: check the required form in the JBI guide for authors]

Problem	Curators need gene–disease relations with evidence, status and source. Extraction tools return only positive pairs, mostly from abstracts.
What is already known	Supervised extractors reach high F1 on in-domain benchmarks such as BioRED. Zero-shot LLMs have been reported to lag in recall and consistency.
What this paper adds	A frozen decision card that gives an LLM high precision, checkable evidence and four-way statuses on three benchmarks and on full text. Evidence that much of the measured gap reflects differing annotation standards. An operating point that recovers recall on held-out data.